SICHUAN UNIVERSITY

四川大学计算机学院（软件学院）

College of Computer Science &College of Software Engineering

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Course: Database Management System

The part I did

SCU library- **Backend & Database Development**

GradeMate- **UI/UX Designing & Frontend Development**

Project 1: SCU Library
Project 2: GradeMate

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SCU Library- A Library Management System

INTRODUCTION:

Background of Library Systems in Universities

In most traditional university settings, library operations are still managed through semi-manual systems or outdated software. This creates inefficiencies when tracking book availability, managing borrow/return records, or analyzing user activity. For students and faculty alike, the lack of a centralized and interactive system often results in delays, miscommunication, and a poor user experience. At Sichuan University, as in many institutions, a digital solution that bridges students, teachers, and the library system could significantly enhance both accessibility and administrative control.

Why I Chose This Project

The inspiration behind this project came from observing the challenges students face when trying to borrow books or check their library history. Sometimes, a student may go to the library only to find the required book unavailable—or even worse, not know which books they've borrowed previously or their return deadlines. I felt that designing a digital library system could eliminate these gaps and offer both students and teachers a smoother, smarter experience. As a Software Engineering student, I also wanted to push my boundaries by combining web technologies, user interface design, and real-time database integration into a full-stack project that is actually useful.

Target Users of the System

This system is designed with three main user groups in mind:

1. **Students**
   - Can register, log in, and access their personal dashboard.
   - Borrow books, check their borrowing history, and leave reviews or feedback on books.
   - Edit their profile and manage their account securely.
2. **Teachers**
   - Use the system similarly to students, with the ability to borrow and review books.
   - May also suggest book additions or track academic reading materials.
3. **Administrators (Library Staff)**
   - The system is structured to support admin roles who can manage inventory, oversee user activity, and generate borrowing reports.

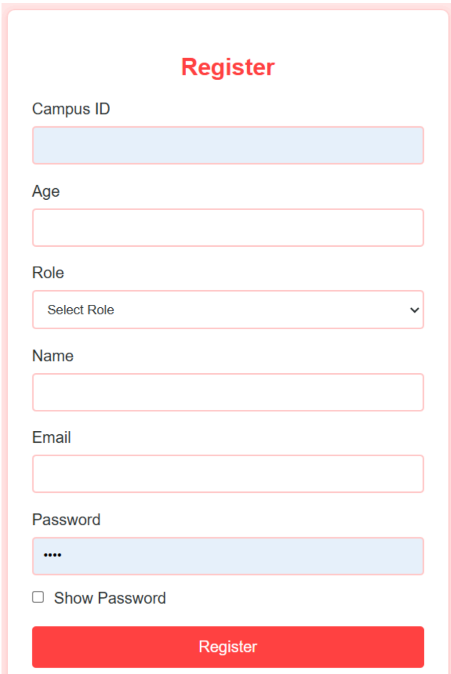
This project is aimed at making library interaction more dynamic, user-centric, and efficient, while laying a foundation for future upgrades like automated due date alerts or book recommendation systems powered by data.

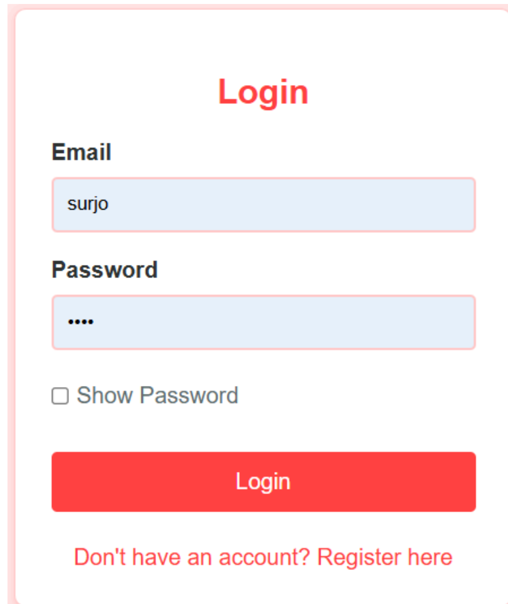
Functional Requirements

The SCU Library Management System includes essential features that ensure seamless user interaction with the database. These core functions allow users to perform basic operations such as account registration, login authentication, and data handling. Below are the key functional components:

1. **User Registration**

New users (students, teachers, others) can create accounts by submitting their campus ID, email, role, name, age, and password through a secure form. Validation checks prevent duplicate entries and ensure correct data input.





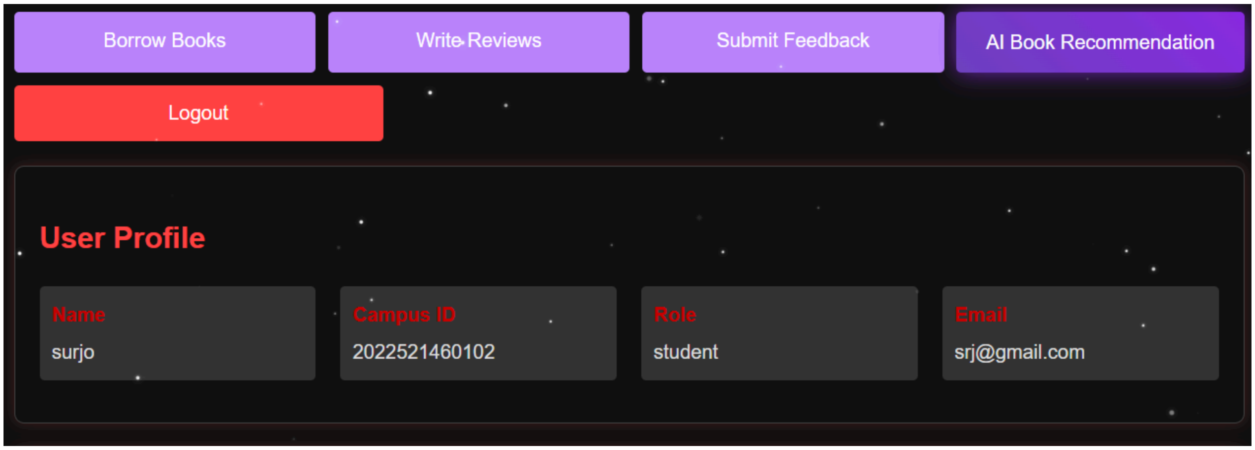
2. **Secure Login & Session Management**

The system uses hashed passwords and session-based login authentication. Upon logging in, users are redirected to a personalized profile/dashboard.

user_id	campus_id	age	role	name	email	password
1	2022521460102	23	student	surjo	srj@gmail.com	$2y$10$d3u83JbKekwbvY087BsYFuecdpXp
2	2022521460205	22	student	lasani	lasani@gmail.com	$2y$10$O65sAE.rP8b5haTQIXL2POdXiML1.
3	54321	32	teacher	Issac	issac@gmail.com	$2y$10$8wqkEBC6MSJH/tuN6Kc4ieGe3/Kek
4	20224329402623	17	others	Shayak Sen	sen@gmail.com	$2y$10$0eeeJC67bnLOri3eGUzCceIrNGFW
5	234567890987654	18	others	Aman Ornob	aman@gmail.com	$2y$10$bZ2yg4uvNOmxf8O5ZfGUPe.A5T7

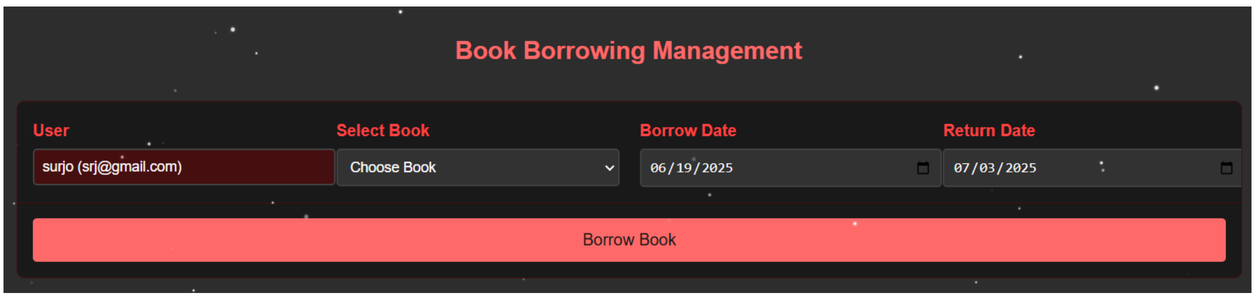
3. **User Dashboard**

Each logged-in user sees a tailored dashboard displaying their details (name, campus ID, role, email) and a history of their borrowed books.



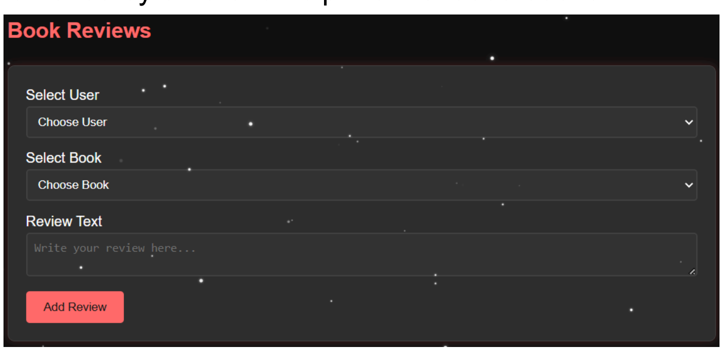
4. **Borrowing System**

Users can request to borrow available books. Borrowed books are recorded with timestamps and tracked in the database.



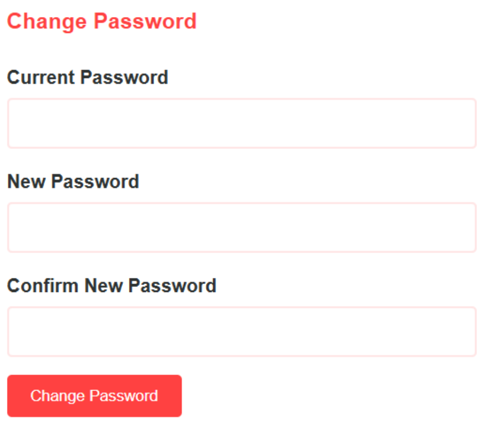
5. **Review & Feedback**

Users can submit feedback or reviews on books they've borrowed. This data is stored and can be retrieved by others to help them choose books.



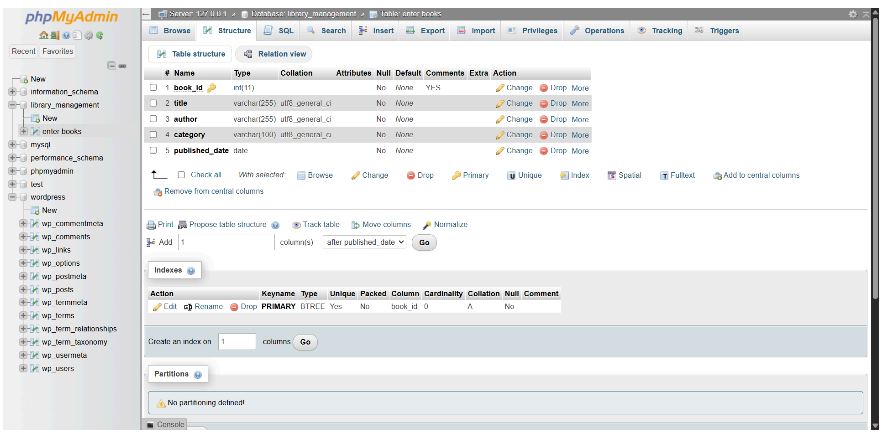
6. **Account Management**

Users can update profile details and change their password through the dashboard, with proper authentication and input validation.

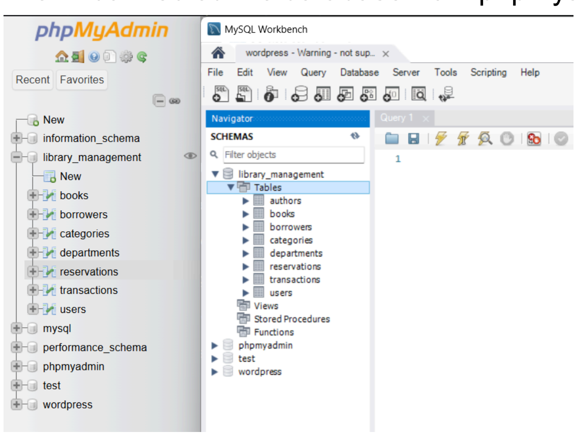


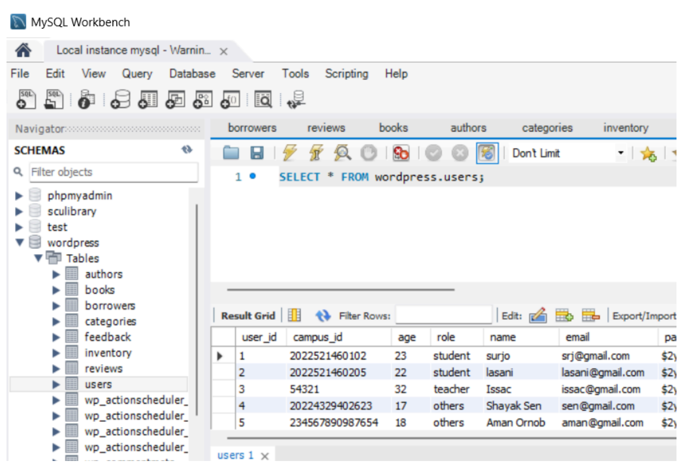
7. Database Sytem

I used PHP MyAdmin for the database management startings.



Then I connected the database from phpmyadmin to my MYSQL Workbench installed in my laptop.





System Planning

Scope:

The SCU Library Management System is designed to help students and teachers manage books, borrowing history, reviews, and feedback in a digital environment with user-friendly access.

Objectives:

- Provide a secure registration and login system.
- Allow users to borrow, review, and give feedback on books.
- Maintain inventory and track borrowing history.

Technology Stack:

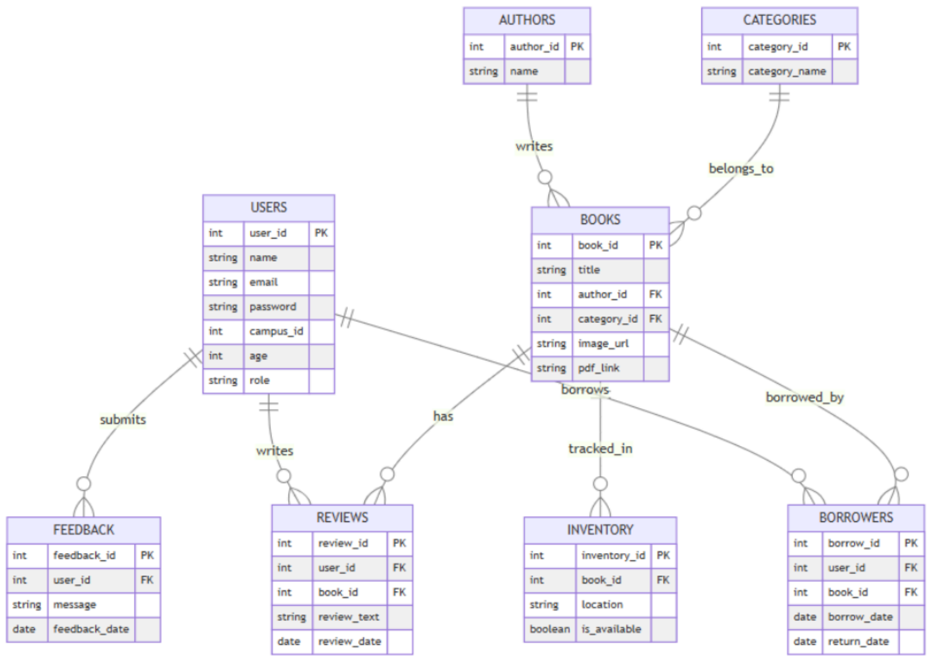
Layer	Tools/Languages Used
Frontend	HTML, CSS, JS, PHP
Backend	PHP,Python
CMS	WordPress + Astra Child
Database	MySQL (accessed via Workbench)
Development	XAMPP (localhost)

Requirement Analysis

User Roles:	Functional Requirements:	Non-functional Requirements:
Students	User registration & login	Secure authentication (password hashing)
Teachers	Book borrowing & return	Responsive interface
Others	Review submission	Fast load time
Admin	Feedback submission	Role-based access control
	Admin control panel	
	Campus ID based user tracking	
	View borrowing history	

Database Design

Entity Relationship Diagram (ERD):



Tables:

users
authors
books
borrowers
feedback
categories
reviews
inventory

Example Table:

```
CREATE TABLE IF NOT EXISTS `authors` (
  `author_id` int(11) NOT NULL AUTO_INCREMENT,
  `name` varchar(100) DEFAULT NULL,
  PRIMARY KEY (`author_id`)
) ENGINE=InnoDB AUTO_INCREMENT=24 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;


CREATE TABLE IF NOT EXISTS `books` (
  `book_id` int(11) NOT NULL AUTO_INCREMENT,
  `title` varchar(100) DEFAULT NULL,
  `image_url` varchar(255) DEFAULT NULL,
  `author_id` int(11) DEFAULT NULL,
  `category_id` int(11) DEFAULT NULL,
  `pdf_link` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`book_id`),
  KEY `author_id` (`author_id`),
  KEY `category_id` (`category_id`)
) ENGINE=InnoDB AUTO_INCREMENT=33 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;


CREATE TABLE IF NOT EXISTS `borrowers` (
```

```
  `borrow_id` int(11) NOT NULL AUTO_INCREMENT,
  `user_id` int(11) DEFAULT NULL,
  `book_id` int(11) DEFAULT NULL,
  `borrow_date` date DEFAULT NULL,
  `return_date` date DEFAULT NULL,
  PRIMARY KEY (`borrow_id`),
  KEY `user_id` (`user_id`),
  KEY `book_id` (`book_id`)
) ENGINE=InnoDB AUTO_INCREMENT=7 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

CREATE TABLE IF NOT EXISTS `categories` (
  `category_id` int(11) NOT NULL AUTO_INCREMENT,
  `category_name` varchar(100) DEFAULT NULL,
  PRIMARY KEY (`category_id`)
) ENGINE=InnoDB AUTO_INCREMENT=20 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;


CREATE TABLE IF NOT EXISTS `feedback` (
  `feedback_id` int(11) NOT NULL AUTO_INCREMENT,
  `message` text DEFAULT NULL,
  `feedback_date` timestamp NOT NULL DEFAULT current_timestamp(),
  PRIMARY KEY (`feedback_id`)
) ENGINE=InnoDB AUTO_INCREMENT=10 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;


CREATE TABLE IF NOT EXISTS `inventory` (
  `inventory_id` int(11) NOT NULL AUTO_INCREMENT,
  `book_id` int(11) DEFAULT NULL,
  `location` varchar(100) DEFAULT NULL,
  `is_available` tinyint(1) DEFAULT NULL,
  PRIMARY KEY (`inventory_id`),
  KEY `book_id` (`book_id`)
) ENGINE=InnoDB AUTO_INCREMENT=36 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;



CREATE TABLE IF NOT EXISTS `reviews` (
  `review_id` int(11) NOT NULL AUTO_INCREMENT,
  `user_id` int(11) DEFAULT NULL,
  `book_id` int(11) DEFAULT NULL,
  `review_text` text DEFAULT NULL,
  `review_date` timestamp NOT NULL DEFAULT current_timestamp(),
  PRIMARY KEY (`review_id`),
  KEY `user_id` (`user_id`),
  KEY `book_id` (`book_id`)
) ENGINE=InnoDB AUTO_INCREMENT=5 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```sql
CREATE TABLE IF NOT EXISTS `users` (
  `user_id` int(11) NOT NULL AUTO_INCREMENT,
  `campus_id` varchar(50) DEFAULT NULL,
  `age` int(11) DEFAULT NULL,
  `role` enum('teacher','student','others') DEFAULT NULL,
  `name` varchar(100) DEFAULT NULL,
  `email` varchar(100) DEFAULT NULL,
  `password` varchar(255) DEFAULT NULL,
  PRIMARY KEY (`user_id`),
  UNIQUE KEY `email` (`email`)
) ENGINE=InnoDB AUTO_INCREMENT=9 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

Alter Tables

```sql
ALTER TABLE `books`
  ADD CONSTRAINT `books_ibfk_1` FOREIGN KEY (`author_id`) REFERENCES `authors` (`author_id`),
  ADD CONSTRAINT `books_ibfk_2` FOREIGN KEY (`category_id`) REFERENCES `categories`
(`category_id`);

 Constraints for table `borrowers
ALTER TABLE `borrowers`
  ADD CONSTRAINT `borrowers_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users` (`user_id`),
  ADD CONSTRAINT `borrowers_ibfk_2` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`);

 Constraints for table `inventory`

ALTER TABLE `inventory`
  ADD CONSTRAINT `inventory_ibfk_1` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`);

 Constraints for table `reviews`

ALTER TABLE `reviews`
  ADD CONSTRAINT `reviews_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users` (`user_id`),
  ADD CONSTRAINT `reviews_ibfk_2` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`);
COMMIT;
```

Normalization:
 All tables are designed in **3rd Normal Form (3NF)** to eliminate redundancy and ensure data integrity.
Constraints:
Primary keys on all tables.
Foreign keys (e.g., `book_id`, `user_id` in `borrowers`, `reviews`).
Custom constraints like email uniqueness and age >= 18. Etc . All table sql files are added to the project Github Link, https://github.com/CodeArtistAladin/DBMS-final-project

Application Design

System Architecture:
A **Client-Server architecture** using WordPress for frontend UI and PHP for backend logic. MySQL serves as the database engine. Data flows from the frontend (HTML/PHP forms) to backend PHP scripts which interact with the database.

Algorithm Implementation:
Password hashing with `password_hash()`
Custom logic to fetch borrowing history and user activity
Basic search & filter features (optional)

Code Snippets

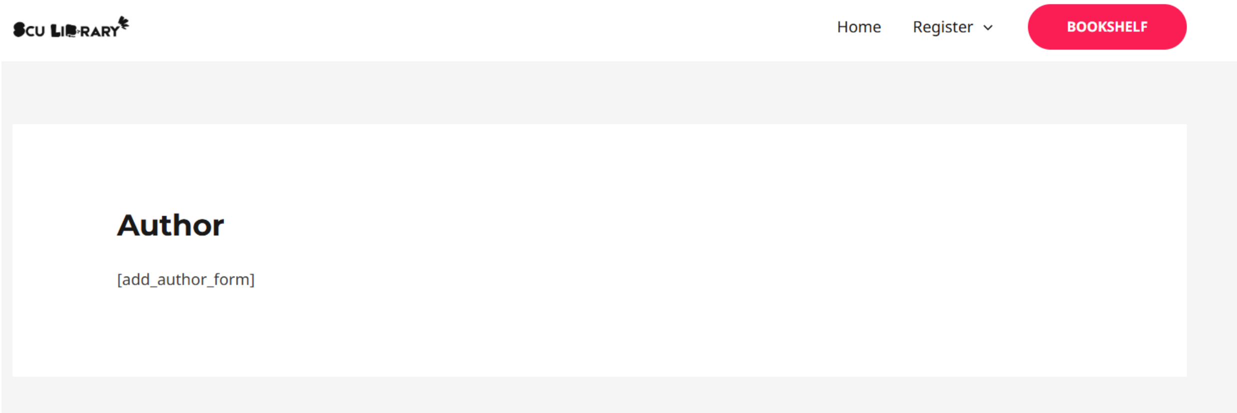
Registration:
```
$password = password_hash($_POST['password'], PASSWORD_BCRYPT);
$sql = "INSERT INTO users (campus_id, age, role, name, email, password)
VALUES (?, ?, ?, ?, ?, ?)";
```
Login:
```
if (password_verify($entered_password, $hashed_password_from_db)) {
$_SESSION['user_id'] = $row['user_id'];
}
```
Profile Fetching:
```
SELECT * FROM borrowers WHERE user_id = $_SESSION['user_id'];
```

Challenges Faced

During the development of the SCU Library system, I encountered several technical and conceptual challenges:

1. **Session Handling in PHP**
Managing user sessions securely and avoiding "headers already sent" errors required careful control of PHP script execution order.

2. **Theme Conflicts in WordPress**
Custom PHP files conflicted with Astra child theme settings, especially for displaying headers and footers. This required working manually in the file system and bypassing some WordPress behaviors.



3. Database Integration

Inserting form data directly into a custom MySQL table (instead of using the default `wp_users`) created issues with linking custom login functionality, which had to be handled using prepared statements and custom session variables.

4. Local Host connectivity

It was a massive headache to keep the XAMPP connected. Apache and Mysql were conflicting with other applications and it was crashing again and again.

```
7:10:39 PM  [Apache]  Error: Apache shutdown unexpectedly.
7:10:39 PM  [Apache]  This may be due to a blocked port, missing dependencies,
7:10:39 PM  [Apache]  improper privileges, a crash, or a shutdown by another method.
7:10:39 PM  [Apache]  Press the Logs button to view error logs and check
7:10:39 PM  [Apache]  the Windows Event Viewer for more clues
7:10:39 PM  [Apache]  If you need more help, copy and post this
7:10:39 PM  [Apache]  entire log window on the forums
```

5. Time Management

Balancing database normalization, front-end design, backend logic, and testing within a fixed schedule was one of the biggest overall challenges.

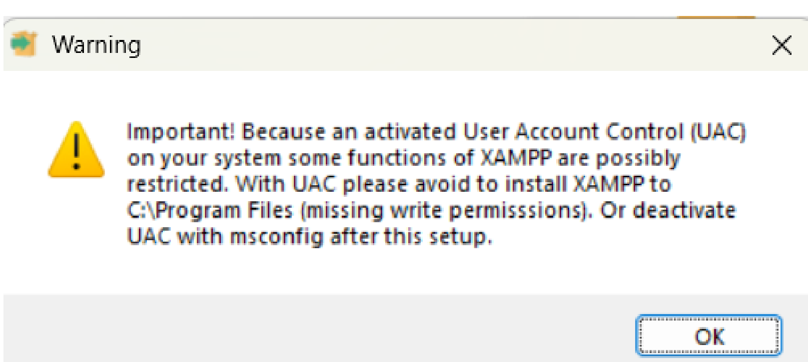
6. Nonce Security

It was another issue that wasted a lot of time to manage or find out the verifying nonce security.

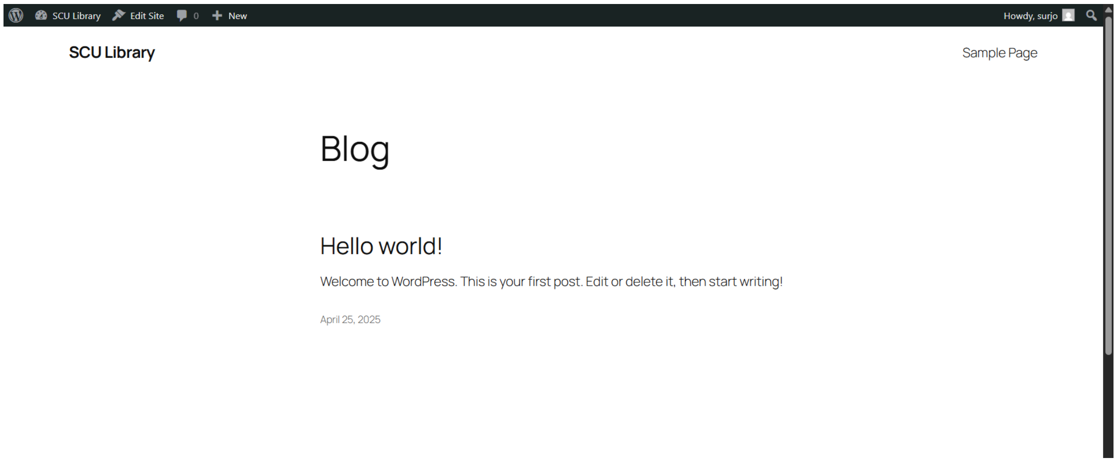
```php
// Verify nonce for security
if (!wp_verify_nonce($_POST['_wpnonce'], 'add_author_action')) {
    wp_die('Security check failed');
}
```

7. XAMPP installation issue

While installing the XAMPP to my pc it was a headache. Here's a screenshot of one,

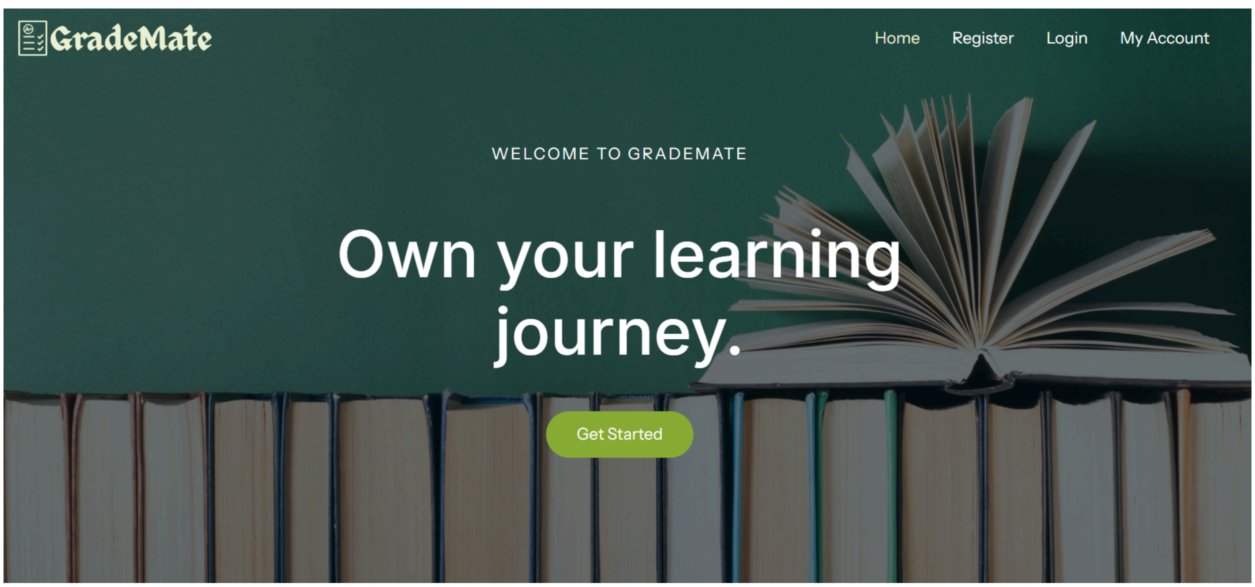


I encountered this pop up. But this didn't cause any issue. So I continue. After installing XAMPP and Wordpress I created the server and launch a website which was looking like this,

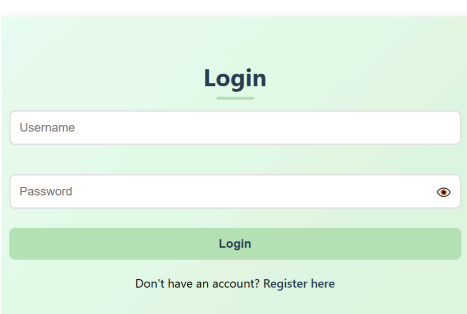


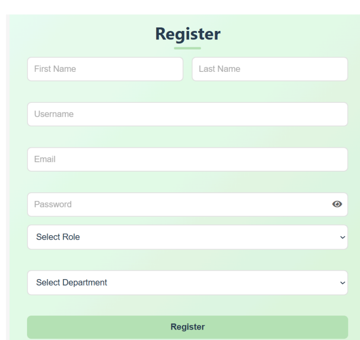
Grade Mate-A Grade Tracking System

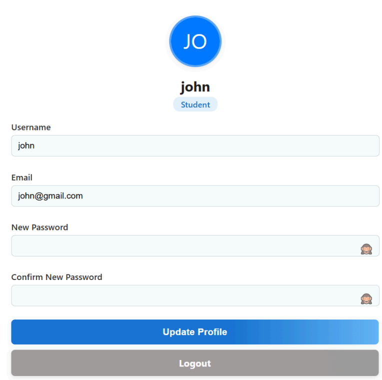
<u>UI Design of GradeMate</u>



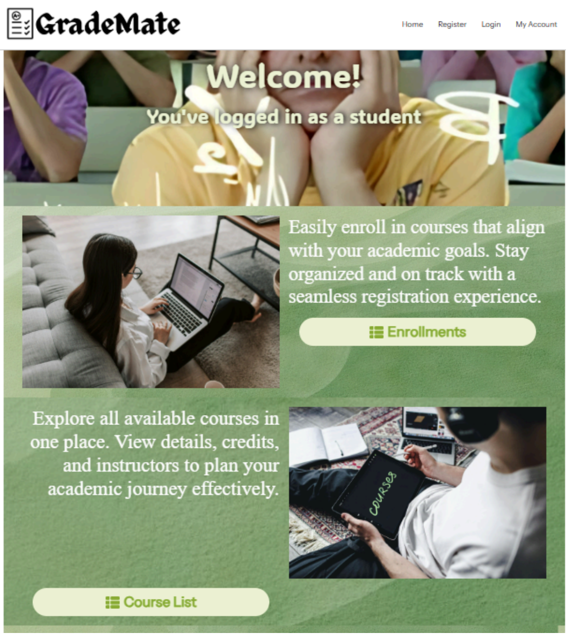
Home Page

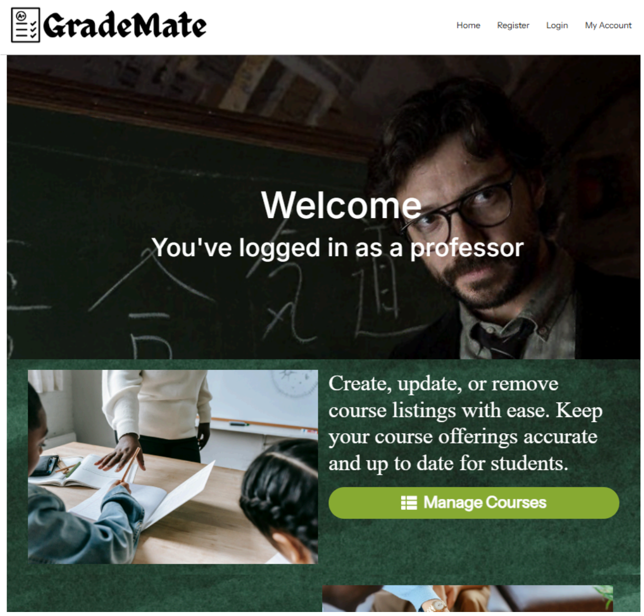




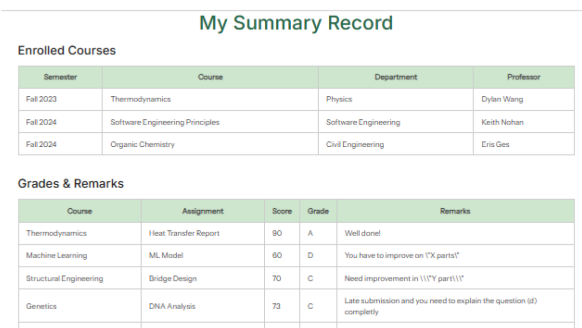


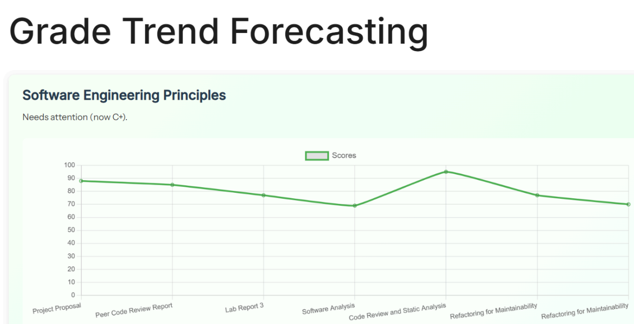
Login **Register** **User Profile**



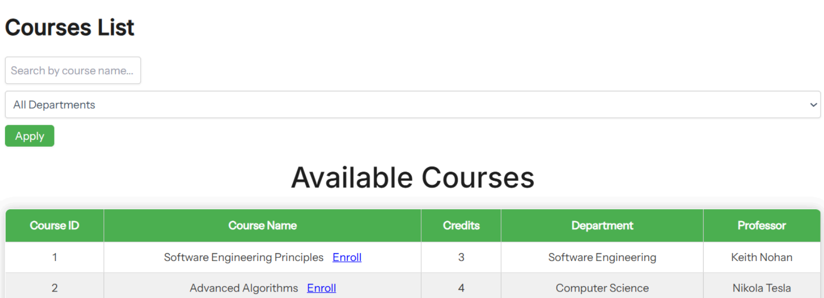


Student Dashboard **Professor dashboard**

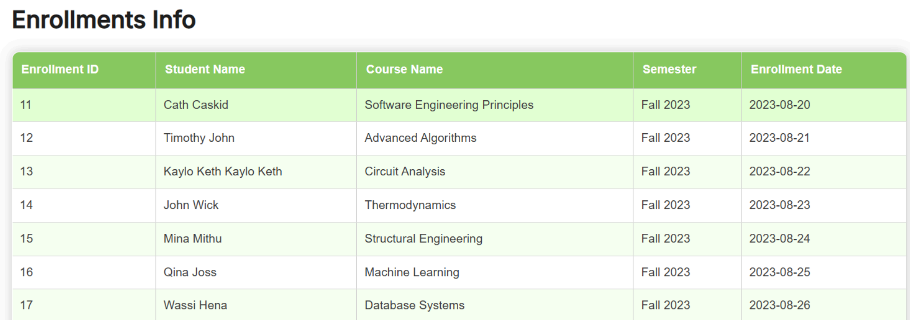




Student Summary Record page **Student Grade Trend Chart**



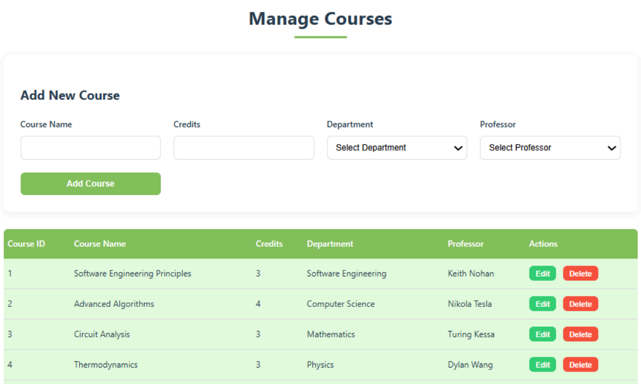
Course Management Page(for Professors)



Enrollments Info

Enrollment ID	Student Name	Course Name	Semester	Enrollment Date
11	Cath Caskid	Software Engineering Principles	Fall 2023	2023-08-20
12	Timothy John	Advanced Algorithms	Fall 2023	2023-08-21
13	Kaylo Keth Kaylo Keth	Circuit Analysis	Fall 2023	2023-08-22
14	John Wick	Thermodynamics	Fall 2023	2023-08-23
15	Mina Mithu	Structural Engineering	Fall 2023	2023-08-24
16	Qina Joss	Machine Learning	Fall 2023	2023-08-25
17	Wassi Hena	Database Systems	Fall 2023	2023-08-26

Enrollment list for both students and professors



Manage Courses

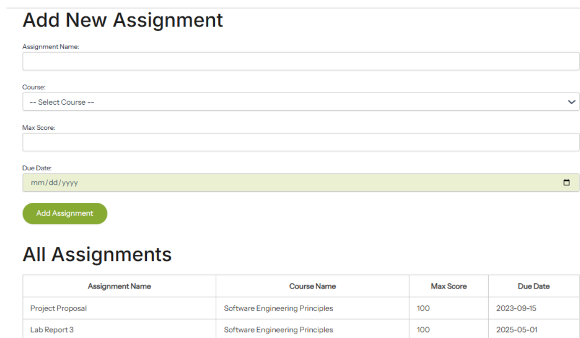
Add New Course

Course Name | Credits | Department: Select Department | Professor: Select Professor

Add Course

Course ID	Course Name	Credits	Department	Professor	Actions
1	Software Engineering Principles	3	Software Engineering	Keith Nohan	Edit Delete
2	Advanced Algorithms	4	Computer Science	Nikola Tesla	Edit Delete
3	Circuit Analysis	3	Mathematics	Turing Kessa	Edit Delete
4	Thermodynamics	3	Physics	Dylan Wang	Edit Delete

Course management for professors



Add New Assignment

Assignment Name:

Course: -- Select Course --

Max Score:

Due Date: mm/dd/yyyy

Add Assignment

All Assignments

Assignment Name	Course Name	Max Score	Due Date
Project Proposal	Software Engineering Principles	100	2023-09-15
Lab Report 3	Software Engineering Principles	100	2025-05-01

Assignment management for professors